

Won Rhim

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<https://github.com/wwrk22>

EXPERIENCE

Hughes Network Systems

Nov 2020 – Nov 2022

Software Engineer

Gaithersburg, MD

- Developed a suite of multi-threaded microservices with C++ that use Protobuf transmissions to build the system controller for a satellite network gateway by performing network resource monitoring and management tasks with 20-millisecond precision.
- Led the design and development of two microservices from the ground up, managing the integration of the services into existing systems by working with owners of other suites of microservices.
- Led the effort to compose the first Helm charts for versioning microservices configuration on Kubernetes clusters, streamlining deployment setup and reducing manual effort for live integration tests.
- Leveraged the C++ template feature to overhaul a network resource statistics management microservice, reducing code length by approximately 10-20%, and enabling reuse of functions for parsing diverse Protobuf messages containing statistics.
- Transitioned microservices suite to an asynchronous messaging architecture using a Publish/Subscribe system with Redis, preventing the loss of at least 20 milliseconds previously spent per UDP message reception.
- Co-authored a Python testing tool to facilitate seamless transition between tests by bringing microservices to a nominal state, eliminating the need for service restarts and decreasing total testing time by 5-10%.
- Initiated collaborative efforts with other engineers to translate hundreds of pages of systems requirements into comprehensive design documents, guaranteeing accuracy through continuous refinement based on acceptance criteria test results.

Fraunhofer CESE

May 2018 – Oct 2018

Software Engineer Intern

Riverdale, MD

- Developed an iOS/Android mobile gaming application for University of Maryland's College of Medicine, Pharmacy, and Nursing students using the MVC pattern, enabling interactive learning experiences on pathogens and diseases.
- Created a hierarchical set of pathogen and disease models using Django's model system, employing Object-Oriented-Programming principles in the design process.
- Teamed with front-end developers to design endpoints and data formats, driving the design and implementation of a back-end REST API to serve pathogen and disease model data.
- Built reusable React Native components that communicated with the application back-end for REST API data.
- Built an application User Interface prototype with React Native that was used as the basis for further UI/UX development.

EDUCATION

George Mason University

May 2020

Bachelor of Science in Computer Science

Fairfax, VA

SKILLS

- **Languages:** C++, C, Ruby, Python, Java, HTML, CSS, JavaScript
- **Tools and Frameworks:** Git, CMake, Boost C++ Libraries, Protobuf, Linux build environment, Agile Methodology, Automated Regression Testing, Microservices, Redis, Helm, Kubernetes, REST API